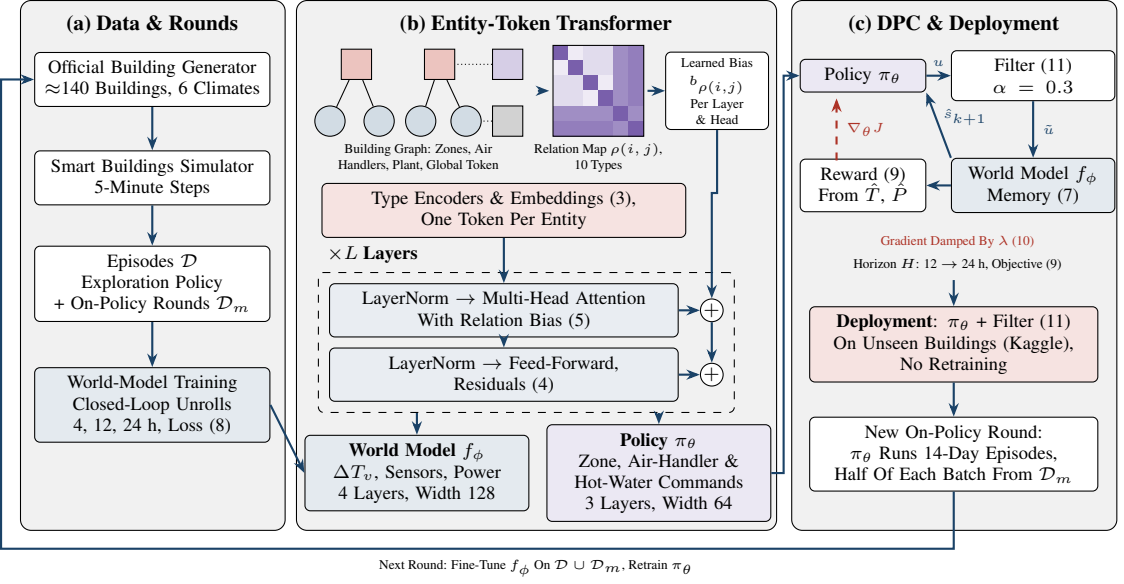

(a) Data & Rounds
Official Building Generator
≈140 Buildings, 6 Climates
Smart Buildings Simulator
5-Minute Steps
Episodes $\mathcal{D}$
Exploration Policy
+ On-Policy Rounds $\mathcal{D}_m$
World-Model Training
Closed-Loop Unrolls
4, 12, 24 h, Loss (8)
(b) Entity-Token Transformer
Building Graph: Zones, Air Handlers, Plant, Global Token
Relation Map $\rho(i,j)$, 10 Types
Learned Bias $b_{\rho(i,j)}$ Per Layer & Head
Type Encoders & Embeddings (3), One Token Per Entity
$\times L$ Layers
LayerNorm → Multi-Head Attention With Relation Bias (5)
LayerNorm → Feed-Forward, Residuals (4)
World Model $f_\phi$
$\Delta T_v$, Sensors, Power
4 Layers, Width 128
Policy $\pi_\theta$
Zone, Air-Handler & Hot-Water Commands
3 Layers, Width 64
(c) DPC & Deployment
Policy $\pi_\theta$
$u$
Filter (11)
$\alpha = 0.3$
$\nabla_\theta J$
$\hat{s}_{k+1}$
$\tilde{u}$
Reward (9)
From $\hat{T}$, $\hat{P}$
World Model $f_\phi$
Memory (7)
Gradient Damped By $\lambda$ (10)
Horizon $H$: 12 → 24 h, Objective (9)
Deployment: $\pi_\theta$ + Filter (11)
On Unseen Buildings (Kaggle), No Retraining
New On-Policy Round:
$\pi_\theta$ Runs 14-Day Episodes,
Half Of Each Batch From $\mathcal{D}_m$
Next Round: Fine-Tune $f_\phi$ On $\mathcal{D} \cup \mathcal{D}_m$, Retrain $\pi_\theta$